



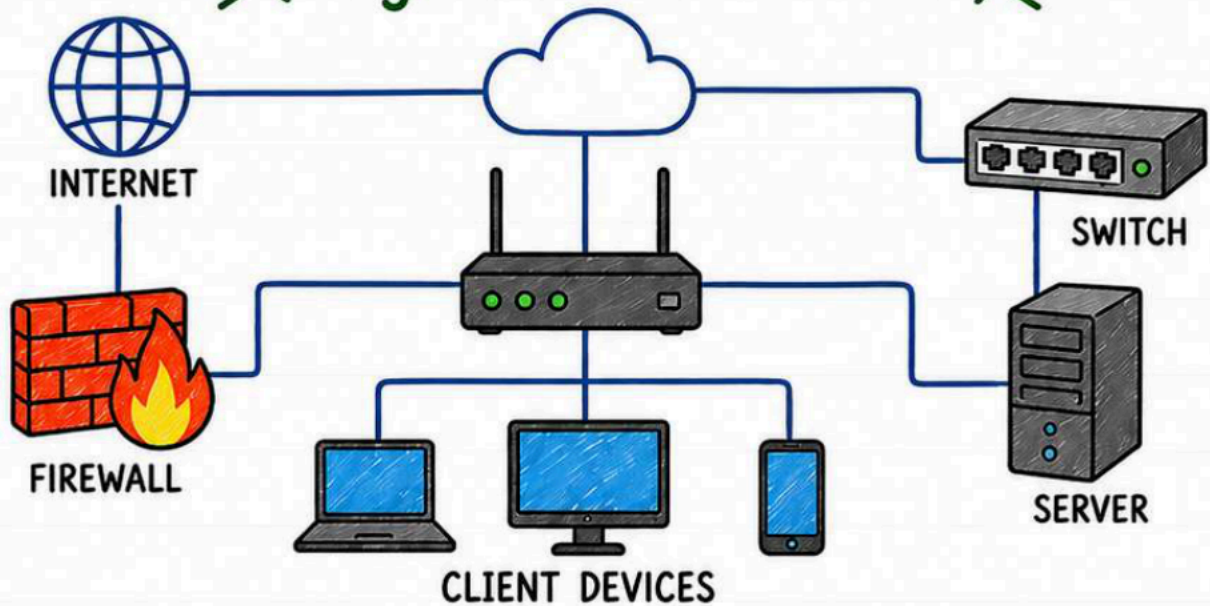
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NETWORKING

COMPLETE NOTES

≡ (Beginner to Advanced) ≡



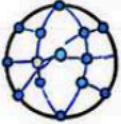
≡ LEARN • UNDERSTAND • APPLY ≡

SWIPE TO NEXT → ≡



Introduction to Computer Networking

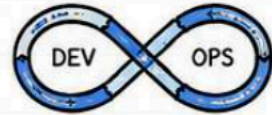
① What is Networking?



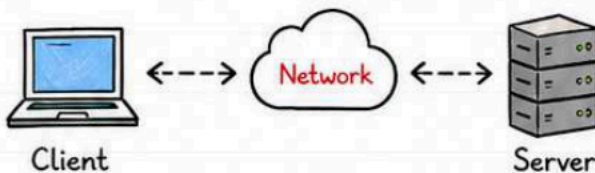
Networking is the process of connecting two or more devices to exchange data and share resources using communication protocols.

② Why Networking matters in DevOps

- ✓ Enables communication between tools, services & users
- ✓ Critical for CI/CD, monitoring, logging & automation
- ✓ Ensures security, performance, scalability & reliability
- ✓ Foundation of Cloud, Containers & Microservices



③ Client ↔ Server Architecture



- Client (User/Device) sends a request.
- Server processes the request.
- Server sends back the response.

④ LAN, WAN & Internet



LAN (Local Area Network)
Connects devices in a small area like home, office, campus.

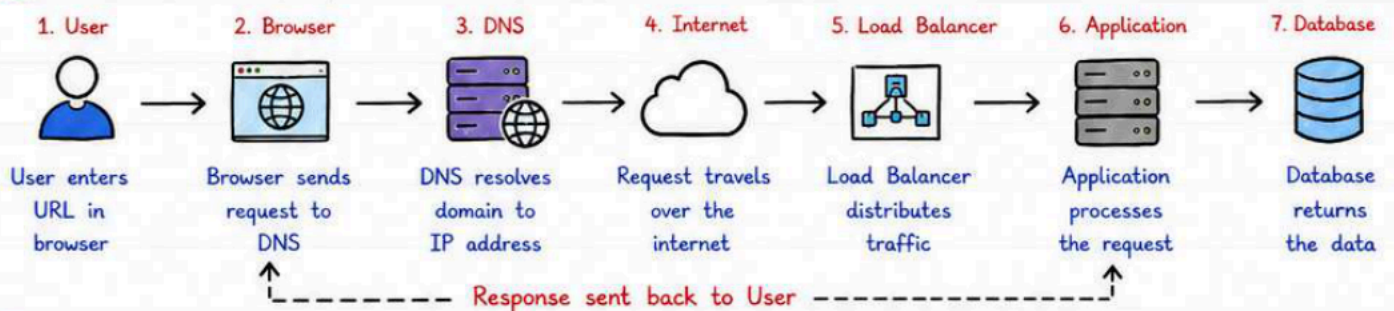


WAN (Wide Area Network)
Connects LANs over large distances like cities, countries.



Internet
Global network of networks that connects the world.

⑤ Real-world Request Flow (Example)



⑥ Where Networking is used in DevOps



Cloud Infrastructure



Containers & Docker



Kubernetes Services



CI/CD Pipelines



Monitoring & Logging



Security & Firewall Rules



Microservices Communication



Interview Tip

Networking is one of the most frequently tested topics in DevOps interviews.



Remember

Every application request depends on networking before reaching the server.



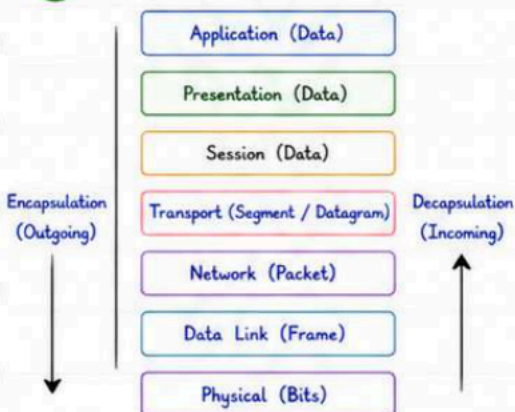
OSI Model (7 Layers)

The OSI (Open Systems Interconnection) model is a conceptual framework that standardizes how data is transmitted over a network.

① 7 Layers Overview

Layer (No.)	Layer Name	Responsibilities	Data Unit (PDU)	Devices at this Layer	Example / Protocols
7 	Application Layer	Provides network services to user applications.	Data	—	HTTP, HTTPS, FTP, SMTP, DNS, SSH, DHCP
6 	Presentation Layer	Handles data format, encryption, compression, translation.	Data	—	SSL/TLS, JPEG, ASCII, MPEG
5 	Session Layer	Manages sessions, establishes, maintains & terminates connections.	Data	—	NetBIOS, RPC, PPTP
4 	Transport Layer	Provides end-to-end delivery, segmentation, flow control, error control.	Segment (TCP) Datagram (UDP)	L4 Switch (Load Balancer)	TCP, UDP
3 	Network Layer	Logical addressing, routing, path selection, packet forwarding.	Packet	Router (L3)	IP (IPv4, IPv6), ICMP, OSPF, BGP
2 	Data Link Layer	Framing, MAC addressing, error detection, access control.	Frame	Switch / Bridge (L2)	Ethernet, PPP, 802.11, MAC
1 	Physical Layer	Transmits raw bits over physical medium.	Bits	Hub, Repeater	Cables, Connectors, Hubs, Signal Standards

② Data Unit (PDU) Flow



Remember

OSI model is a reference model.
Real networks use TCP/IP model in practice.

③ Easy Memory Trick

Please Do Not
Throw Sausage
Pizza Away

P - Physical
D - Data Link
N - Network
T - Transport
S - Session
P - Presentation
A - Application

④ Key Takeaways

- ✓ Upper 3 layers (5-7) focus on application communication.
- ✓ Middle layer (4) ensures reliable end-to-end delivery.
- ✓ Lower 2 layers (1-2) handle actual data transmission.
- ✓ Data flows down from Layer 7 to 1 when sending and up from 1 to 7 when receiving.



Interview Tip

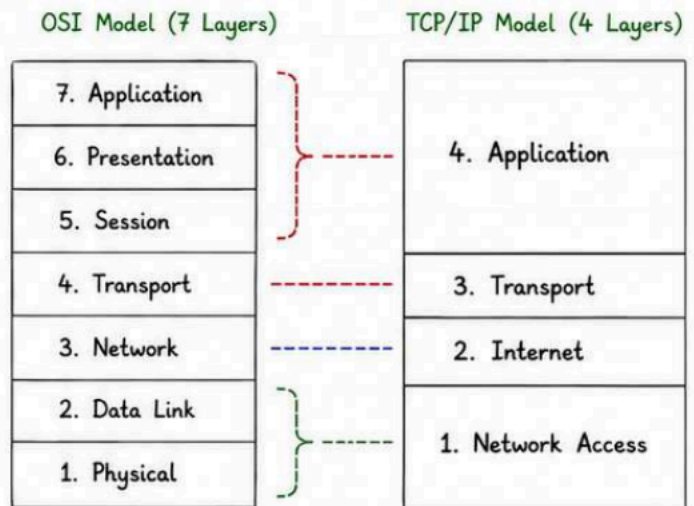
Be ready to explain each layer easily with examples and devices. Remember the memory trick under pressure!

TCP/IP Model (4 Layers)

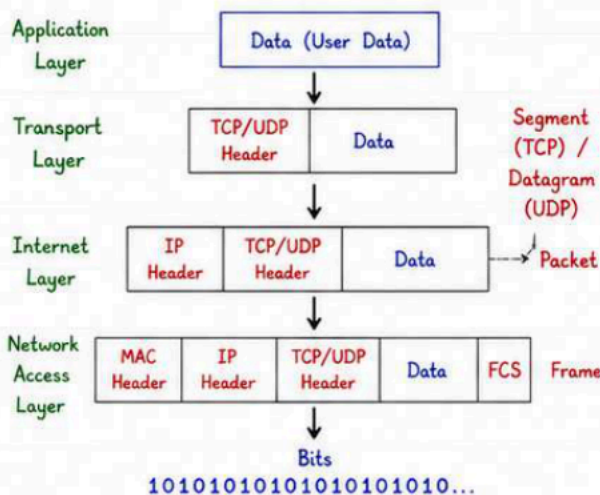
① 4 Layers of TCP/IP Model

Layer (Top to Bottom)		Key Responsibilities	Example Protocols
4	Application Layer	- Provides network services to applications. - Handles user interfaces, APIs, data formatting.	HTTP, HTTPS, FTP, DNS, SSH, SMTP, DHCP
3	Transport Layer	- End-to-end communication. - Segmentation, reliability, port management.	TCP, UDP
2	Internet Layer	Logical addressing and routing of packets across networks.	IP (IPv4, IPv6), ICMP, ARP
1	Network Access Layer	- Transmits data on the physical network. - Framing, MAC addressing.	Ethernet, Wi-Fi, PPP

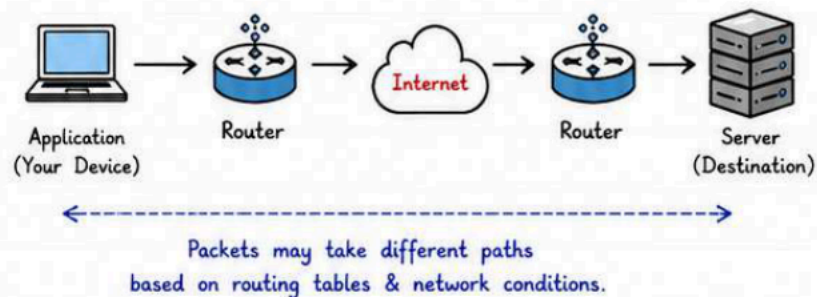
② OSI vs TCP/IP Model



③ Encapsulation (At Sender)



④ Packet Journey (Sender to Receiver)



Remember

Each layer adds its own header (and sometimes trailer) while sending, and removes it while receiving.

⑤ Why TCP/IP Model is Used?

- ✓ Simple and easy to understand.
- ✓ Widely adopted across the Internet and all modern networks.
- ✓ Supports interoperability between different vendors and systems.
- ✓ Efficient, scalable and works well for large distributed systems.
- ✓ Forms the foundation for protocols used in DevOps tools, Cloud, Containers, and Microservices.

 Interview Tip

In interviews, focus on understanding layer functions, real-world examples and how data moves from one layer to another.



IP Addressing

① IPv4 vs IPv6

IPv4	IPv6
32-bit address - Format: a.b.c.d (0 - 255 each octet) - Example: 192.168.1.10 - Total Addresses: ~4.3 Billion	128-bit address - Format: xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx - Example: 2001:0db8:85a3:8a2e:0370:7334 - Total Addresses: ~340 Undecillion



IPv6 was created to overcome IPv4 address exhaustion and provide better scalability & security.

② Public vs Private IP



Public IP (Internet Reachable)

- Assigned by ISP
- Unique globally
- Used to communicate over the Internet
- Example: 8.8.8.8



Private IP (Internal Use)

- Used inside private networks (LAN/VPC)
- Not routable on the Internet
- Saves public IPs

Examples: 10.0.0.0/8 172.16.0.0/12 192.168.0.0/16

③ Static vs Dynamic IP

Static IP

- Manually configured
- Does not change
- Used for servers, devices that need consistent access



Example: 49.36.12.5

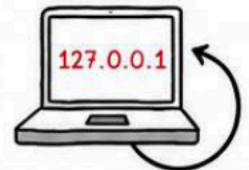
Dynamic IP

- Assigned automatically by DHCP
- Can change over time
- Used for client devices

Example: QA Insights Developer Laptop (Changes periodically)

④ Loopback Address

- Used by a device to communicate with itself.
- Used for testing and local development.
- IPv4 Loopback: 127.0.0.1
- IPv6 Loopback: ::1



Packets sent to loopback never leave the device.

⑤ CIDR (Classless Inter-Domain Routing) - Introduction

- CIDR is a method of allocating IP addresses and routing IP packets.
- It replaces traditional classful addressing.
- Notation: IP Address / Prefix Length
- Example: 192.168.1.0/24
 - ✓ 24 means first 24 bits are network part
 - ✓ Remaining 8 bits are for hosts
- CIDR helps in efficient IP usage and route summarization.

192.168.1.0/24

Network (24 bits)	Host (8 bits)
192 . 168 . 1	0

Supports $2^8 - 2 = 254$ usable hosts

Common CIDR Ranges			
/8	→	16,777,214	hosts
/16	→	65,534	hosts
/24	→	254	hosts
/30	→	2	hosts (Point-to-Point)



Interview Tip

Understand IP types, ranges, and CIDR deeply - these are frequently asked in DevOps & Networking interviews!



Remember

IP addressing is the foundation of all networking. Without IP, devices cannot communicate!

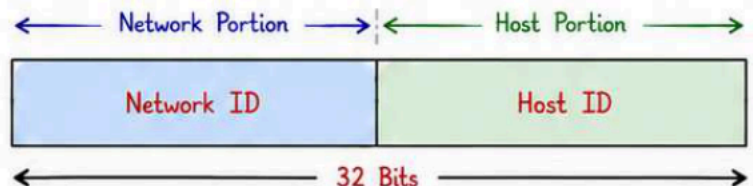


Subnetting & CIDR

① Key Terms

- **Network ID** : Identifies the network portion of an IP.
- **Host ID** : Identifies the specific device in that network.
- **Subnet Mask** : Divides IP into network and host portions.
- **CIDR Notation** : Short and efficient way to represent IP and subnet mask.
- **Available Hosts** : Total usable IP addresses in a subnet.

② IP Address Structure (IPv4)



Example:

IP Address : 192.168.1.10 Subnet Mask : 255.255.255.0 (/24)

IP Address : 11000000. 10101000. 00000001. 00001010

Subnet Mask : 11111111. 11111111. 11111111. 00000000

Network ID
(24 bits)

Host ID
(8 bits)

③ Subnet Mask

A subnet mask is a 32-bit number that tells which part of the IP is Network ID and which part is Host ID.

Common Subnet Masks

CIDR	Subnet Mask	Network / Host Bits
/8	255.0.0.0	8 / 24
/16	255.255.0.0	16 / 16
/24	255.255.255.0	24 / 8
/25	255.255.255.128	25 / 7
/26	255.255.255.192	26 / 6
/27	255.255.255.224	27 / 5
/28	255.255.255.240	28 / 4
/30	255.255.255.252	30 / 2

④ CIDR Notation

CIDR (Classless Inter-Domain Routing) removes the need for classful networks and allows variable length subnetting.

Format: IP Address / Prefix Length

Examples:

- 192.168.1.0/24 → First 24 bits are network portion
- 10.0.0.0/8 → First 8 bits are network portion
- 172.16.5.0/16 → First 16 bits are network portion



Interview Tip

CIDR helps in efficient IP allocation, route summarization and reduces routing table size in Routers.

⑤ Available Hosts Calculation

Example Calculation

Given IP : 192.168.1.10/24

Subnet Mask : 255.255.255.0

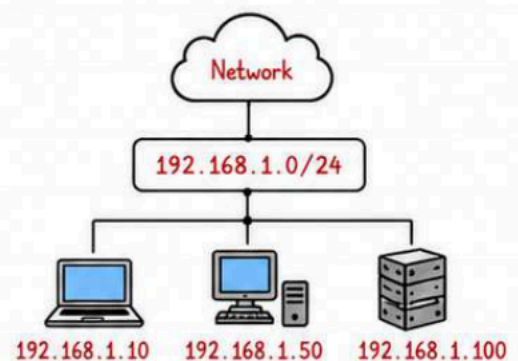
Formula:

$$\text{Total Hosts} = 2^n$$

$$\text{Usable Hosts} = 2^n - 2$$

Where 'n' = number of host bits
(We subtract 2 for Network ID and Broadcast ID)

CIDR	/24
Host Bits (n)	8
Total Hosts (2^n)	$2^8 = 256$
Usable Hosts ($2^n - 2$)	$256 - 2 = 254$
Host Range	192.168.1.1 - 192.168.1.254
Network ID	192.168.1.0
Broadcast ID	192.168.1.255



Remember

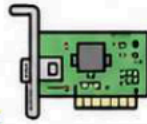
- ✓ Network ID → First IP in the subnet (all host bits = 0)
- ✓ Broadcast ID → Last IP in the subnet (all host bits = 1)



MAC Address & ARP

① MAC Address (Media Access Control)

- A unique hardware address assigned to Network Interface Card (NIC).
- Works at Data Link Layer (Layer 2).
- 48-bit address (6 groups of 2 hex digits).
- Format: 00:1A:2B:3C:4D:5E
- Uniquely identifies a device on a local network.



MAC Example
00:1A:2B:3C:4D:5E

② Physical vs Logical Address

Physical Address (MAC)

- Hardware address
- Layer 2 (Data Link)
- Assigned by Manufacturer
- Used for local network communication

Example:

00:1A:2B:3C:4D:5E

Logical Address (IP)

- Software address
- Layer 3 (Network)
- Assigned manually or by DHCP
- Used for end-to-end communication

Example:

192.168.1.10

③ ARP (Address Resolution Protocol)

- ARP maps IP address (Logical) to MAC address (Physical).
- Works within the local network.
- If sender doesn't know the MAC of destination IP, it sends an ARP Request (Broadcast).
- Destination replies with ARP Reply (Unicast) containing its MAC.

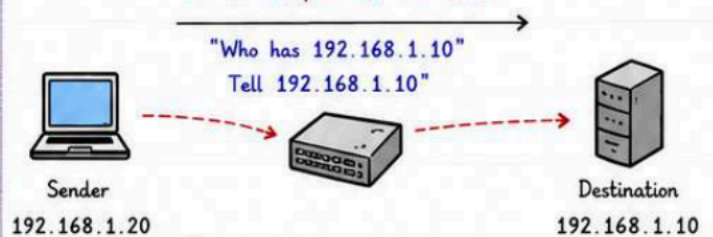


ARP helps find the MAC address of a device using its IP.

Remember
ARP is like asking:
"Who has 192.168.1.10"
Tell 192.168.1.10 that
I am 192.168.1.20."

④ ARP Process (Broadcast & Reply)

1. ARP Request (Broadcast)



2. ARP Reply (Unicast)

"192.168.1.10 is at
00:1A:2B:3C:4D:5E"

⑤ Switch MAC Table

- Switch learns MAC addresses from incoming frames.
- Stores in MAC Address Table with port information.
- Used to forward frames only to the correct port (not broadcast).

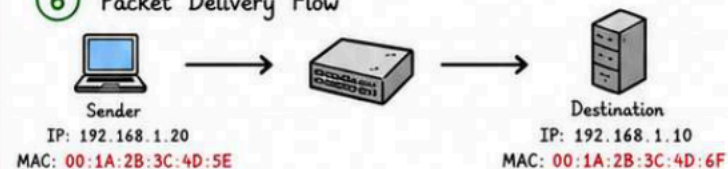
Example MAC Table

MAC Address	Port
00:1A:2B:3C:4D:5E	Port 1
00:1A:2B:3C:4D:6F	Port 2
00:1A:2B:3C:4D:7A	Port 3
00:1A:2B:3C:4D:8B	Port 4
...	...

Key Points

- ✓ MAC = Hardware Address (Layer 2)
- ✓ IP = Logical Address (Layer 3)
- ✓ ARP maps IP → MAC
- ✓ Switch uses MAC Table for intelligent forwarding

⑥ Packet Delivery Flow



1. Sender checks if destination IP is in same subnet.
2. Uses ARP to get destination MAC.
3. Sends frame to Switch (with destination MAC).
4. Switch looks up MAC Table and forwards to correct port.
5. Destination receives the frame and processes it.

★ Interview Tip

Be clear about the difference between MAC & IP, how ARP works, and how switches learn & forward frames. These are commonly asked in DevOps, Network & Cloud interviews.



TCP vs UDP

TCP and UDP are Transport Layer protocols used to transmit data between applications over an IP network.

① Connection-Oriented vs Connectionless

• TCP is Connection-Oriented

A connection is established before data transfer and closed after completion.

• UDP is Connectionless

No connection setup. Data is sent directly without establishing a session.

③ Reliability vs Speed

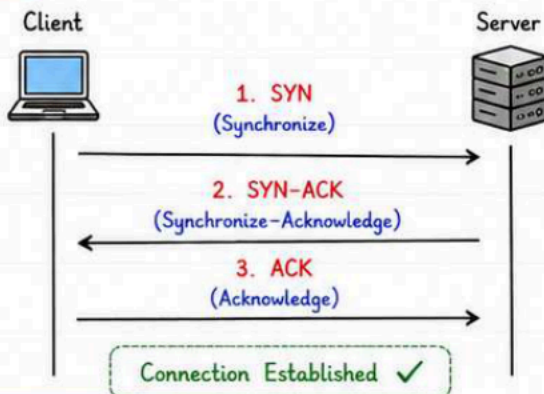
TCP

- ✓ Reliable delivery
- ✓ Ensures data arrives in order
- ✓ Error checking & retransmission
- ✓ Slower due to overhead

UDP

- ✓ Best-effort delivery
- ✓ No guarantees of order or delivery
- ✓ No retransmission or error recovery
- ✓ Faster and lightweight

② Three-Way Handshake (TCP)



④ Real-World Examples

TCP (Reliable Apps)

- Web Browsing (HTTP/HTTPS)
- Email (SMTP, IMAP, POP3)
- File Transfer (FTP, SFTP)
- Database Connections (MySQL, PostgreSQL)

UDP (Fast Apps)

- Online Gaming
- Video Streaming (Live)
- Voice over IP (VoIP)
- DNS Queries

⑤ TCP vs UDP Comparison

Feature	TCP	UDP
Type	Connection-Oriented	Connectionless
Connection Setup	Required (Three-way Handshake)	Not Required
Reliability	Reliable (ACK + Retransmission)	Best-effort (No guarantees)
Ordering	Maintains Order	No Order Guarantee
Error Checking	Yes (Checksum, ACK)	Yes (Checksum only)
Data Delivery	Guaranteed	Not Guaranteed
Speed	Slower	Faster
Overhead	Higher (Header + Handshake)	Lower (Minimal Header)
Data Type	Stream (Continuous)	Datagram (Independent packets)
Header Size	20-60 Bytes	8 Bytes
Use Cases	Web, Email, File Transfer, DB, SSH	Gaming, Streaming, DNS, VoIP, IoT



Interview Tip

If data loss is acceptable but speed is critical → use UDP.

If accuracy and reliability are important → use TCP.



Remember

TCP = Trust, Confirm, Protect

UDP = User, Datagram, Performance



Key Takeaway

TCP is like a phone call (connection, reliable, ordered).

UDP is like a walkie-talkie (no setup, fast, may lose packets).



VS





Ports & Common Protocols

Ports are logical endpoints used by protocols to identify applications and services on a host.

Protocol / Service	Port	Transport	Purpose
 HTTP (HyperText Transfer Protocol)	80	TCP	Unencrypted web traffic
 HTTPS (HTTP Secure)	443	TCP	Encrypted web traffic (SSL/TLS)
 SSH (Secure Shell)	22	TCP	Secure remote login & command execution
 DNS (Domain Name System)	53	UDP/TCP	Domain name resolution
 FTP (File Transfer Protocol)	21 (Control) 20 (Data)	TCP	File transfer (unencrypted)
 SFTP (SSH File Transfer Protocol)	22	TCP	Secure file transfer over SSH
 SMTP (Simple Mail Transfer Protocol)	25	TCP	Sending emails
Database Ports			
 MySQL	3306	TCP	MySQL Database
 PostgreSQL	5432	TCP	PostgreSQL Database
 MongoDB	27017	TCP	MongoDB Database
 Redis	6379	TCP	Redis Cache / Key-Value Store



Key Points

- ✓ Port + IP address uniquely identifies a service.
- ✓ Well-known ports : 0 - 1023
- ✓ Registered ports : 1024 - 49151
- ✓ Dynamic / Private ports : 49152 - 65535



Interview Tip

Always mention the protocol along with the port (e.g., 80/TCP, 53/UDP).
Firewall rules are based on IP, port and protocol.